Constructing and questioning an ID3 tree

Worked entropy calculation, minimality, irrelevant attributes and unseen-case counterexamples.

Preliminaries

A categorical decision tree asks feature-value questions and returns a class label at a leaf. Its **size** will mean the number of internal decision nodes (questions), not the total number of nodes. We work with the same eight synthetic outdoor-activity examples as in Week 3 Course and Laboratory.

#	Sky	Wind	Temp.	Go?
1	sunny	calm	warm	yes
2	sunny	calm	cold	yes
3	sunny	windy	warm	yes
4	sunny	windy	cold	no
5–8	rainy	both	both	no

The final row stands for all four combinations of wind and temperature when sky is rainy. The dataset contains three yes and five no labels. Use the convention $0 \log_2 0 = 0$.

Worked Example A — Choose the root question

Entropy with probabilities p_i is $H = -\Sigma\, p_i \log_2 p_i$. Information gain is the parent entropy minus the average entropy of the subsets created by a question.

$$H(\text{root}) = -(3/8)\log_2(3/8) - (5/8)\log_2(5/8) \approx 0.9544.$$

Splitting on sky gives four sunny cases (three yes, one no) and four rainy cases (all no). Therefore

$$H(\text{after sky}) = (4/8)\cdot 0.8113 + (4/8)\cdot 0 = 0.4056.$$

$$\text{Gain(sky)} = 0.9544 - 0.4056 = \mathbf{0.5488\ bits.}$$

This is the best candidate root. In Problem 1, you must also justify why both alternatives are weaker.

Problem 1 — Compare candidate questions

a. For wind = calm and wind = windy, count the yes and no labels.

b. Calculate the entropy of both groups and the weighted entropy after splitting on wind.

c. Repeat for temperature = warm/cold, then compare all three information gains.

> **Selected result.** For wind, the two subsets contain (2 yes, 2 no) and (1 yes, 3 no). The weighted entropy is (1 + 0.8113)/2 = 0.9056, so the gain is approximately **0.0488**. Temperature has the same count distribution and gain. The sky gain 0.5488 is strictly larger.

Problem 2 — Finish the tree

a. Explain why the rainy branch can immediately become a no leaf.

b. Within the sunny branch, calculate the gain of wind and temperature.

c. If a tie is resolved in favour of wind, draw the complete tree.

d. Trace all eight training examples through it and count the decision nodes.

Hint: The entropy of the sunny branch is about 0.8113. Splitting it on wind leaves a pure two-example group and a mixed two-example group, resulting in remaining entropy 0.5.

> **Selected result.** Gain within the sunny branch is approximately 0.3113 for either remaining feature. One valid tree uses three internal decision nodes:
>
> ```
> if sky == "rainy": return "no"
> if wind == "calm": return "yes"
> if temperature == "warm": return "yes"
> return "no"
> ```

Problem 3 — Is the tree minimal?

Assume each internal node may ask only about one of sky, wind and temperature, and may branch on its observed categorical values.

a. For each feature, find two records which differ only in that feature and have opposite labels.

b. Conclude that the target function genuinely depends on all three features.

c. Show that no tree with fewer than three internal decision nodes can represent the entire eight-row truth table exactly.

> **Selected proof.** Sky matters for fixed calm/cold: sunny → yes, rainy → no. Wind matters for sunny/cold: calm → yes, windy → no. Temperature matters for sunny/windy: warm → yes, cold → no. A tree with at most two decision nodes could ask about at most two distinct features, so at least one relevant feature could never influence its answer. It would be incorrect on a pair of the above witnesses. The construction in Problem 2 achieves three nodes, so **three is necessary and sufficient.**

Problem 4 — An irrelevant additional attribute

Introduce a fourth Boolean feature, indoor-heating, but suppose it never affects the desired outdoor label once the other three features are known.

a. Explain how a tree that unnecessarily branches on indoor-heating can be simplified without changing its prediction for any input.
b. Why does the claim require the feature to be truly irrelevant on the *entire input domain*, not merely on the observed examples?

Hint: If the output is independent of this extra feature, replacing every test of it consistently by the same fixed branch preserves the target function.

Problem 5 — Incomplete data and a counterexample

Remove only record 4 (sunny, windy, cold → no) from the training examples.

a. Construct one tree consistent with all seven remaining rows that predicts no for the missing row.
b. Construct another tree consistent with the same seven rows that predicts yes for that row.
c. Explain why 100% training accuracy on seven examples is insufficient to choose between these models.

Hint: The second tree may simply decide yes whenever sky is sunny. Every remaining sunny example has label yes.

Final observation — Greedy is not globally optimal by definition

ID3 chooses the feature with maximum information gain at the current node. Its definition does not compare every possible completed decision tree, so **local gain does not by itself prove a globally minimum-size tree in general**. In this particular eight-row example, however, Problem 3 provides a separate minimality proof.

Optional computational assistant: Write a short Python loop to enumerate and compare the eight input combinations, but explain the minimality argument in words. No Dafny, proof assistant or theorem prover is required.

References: IOAI classical-ML syllabus · Experience AI — Decision Trees.